

PF-04: Personal Website & DNS Walkthrough

LIVE URL: <https://abdulsamiuthwal-portfolio.vercel.app/> | AUTHOR: Abdul Sami | TRACK: General AI Fluency

1. What is DNS? (The Internet's Phonebook)

Computers communicate using numerical IP addresses (e.g. 76.76.21.21). **DNS (Domain Name System)** acts as an automated internet phonebook that translates human-friendly domain names (abdulsamiuthwal-portfolio.vercel.app) into numerical IP addresses so browsers can load content.

2. What is a CNAME Record? (The Alias Pointer)

A **CNAME (Canonical Name) Record** maps one domain alias to another domain name instead of a static IP address.

- **Example Mapping:** abdulsamiuthwal.flyrank.ai → abdulsamiuthwal-portfolio.vercel.app.
- **Why it matters:** Host servers dynamically balance traffic across IP clusters. CNAME records ensure custom domains remain connected automatically without manual IP updates.

3. Step-by-Step Resolution Flow (Request Lifecycle)

1. **User Request:** User types <https://abdulsamiuthwal-portfolio.vercel.app/> in Chrome.
2. **Resolver Query:** Browser asks local ISP Recursive Resolver for host IP.
3. **Nameserver Hierarchy:** Resolver queries Root Nameserver → .app TLD Nameserver → Vercel Authoritative Nameserver.
4. **Response & HTTPS Handshake:** Authoritative server returns host IP (76.76.21.21). Browser verifies SSL certificate padlock and renders portfolio.

4. Capstone Subdomain Provisioning Checklist (yourname.flyrank.ai)

1. **Ops Provisioning:** FlyRank Ops creates CNAME record mapping abdulsamiuthwal.flyrank.ai → Vercel live target.
2. **Host Registration:** Add abdulsamiuthwal.flyrank.ai under Vercel Custom Domains.
3. **Global Propagation:** Wait 2–5 minutes for global DNS caches to refresh.
4. **Padlock Verification:** Confirm HTTPS SSL certificate active on <https://abdulsamiuthwal.flyrank.ai>.